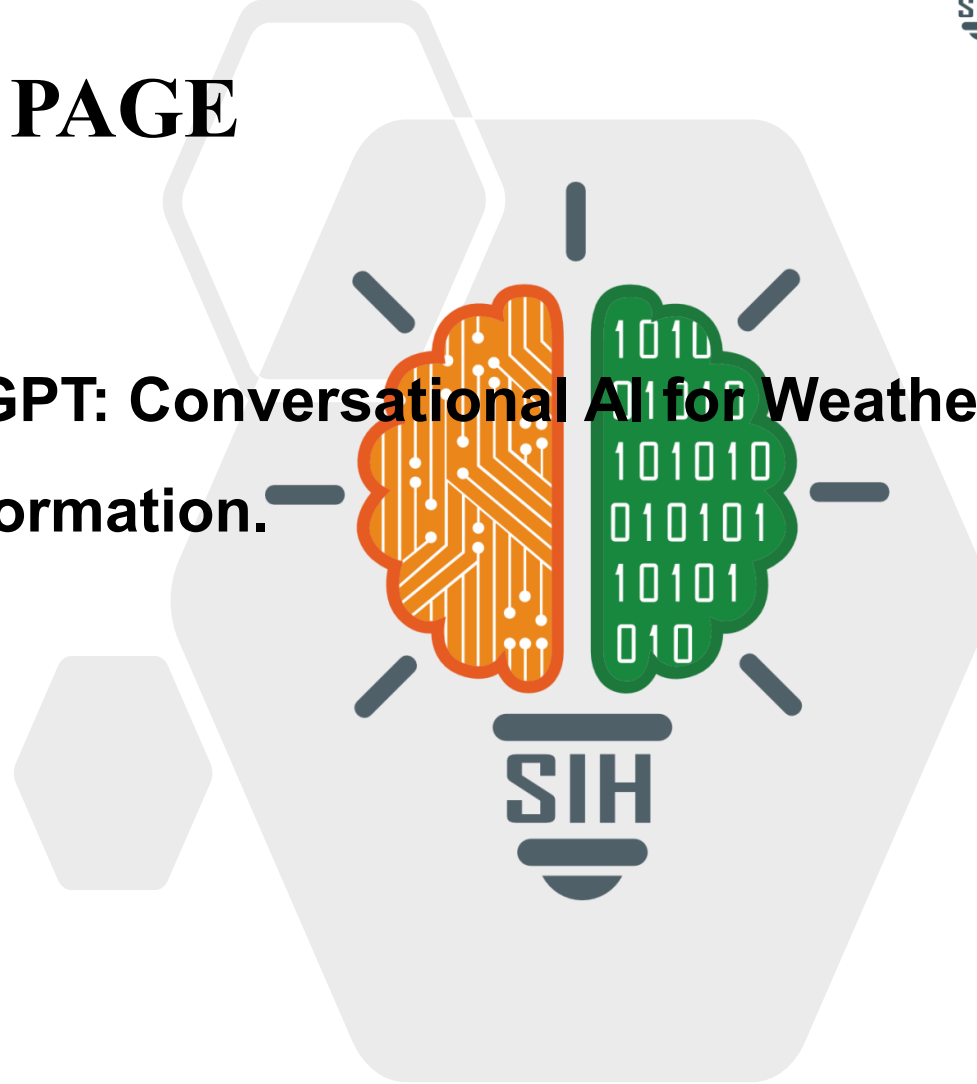


SMART INDIA HACKATHON 2026



TITLE PAGE

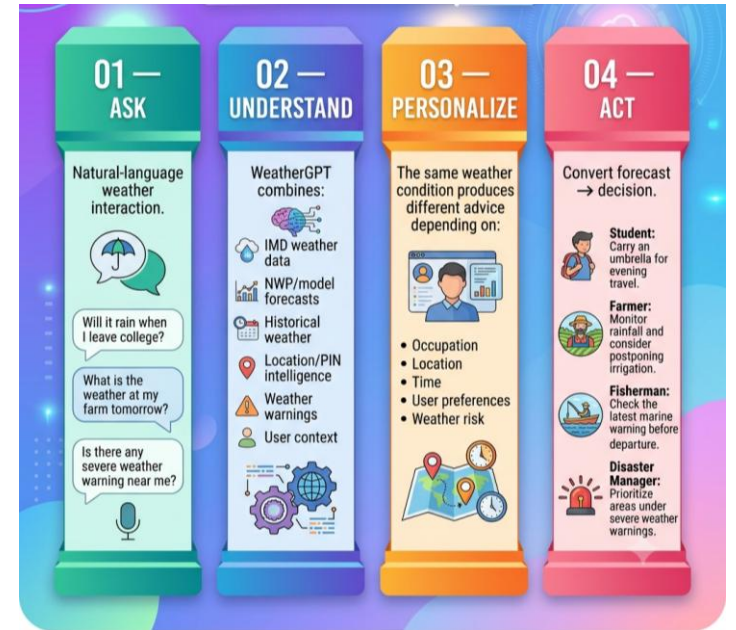
- **Problem Statement ID – 26068**
- **Problem Statement Title – WeatherGPT: Conversational AI for Weather Forecasting, Alerts, and Climate Information.**
- **Theme – Disaster Management**
- **PS Category – Software**
- **Team ID –**
- **Team Name – Binary Beasts**



IDEA TITLE

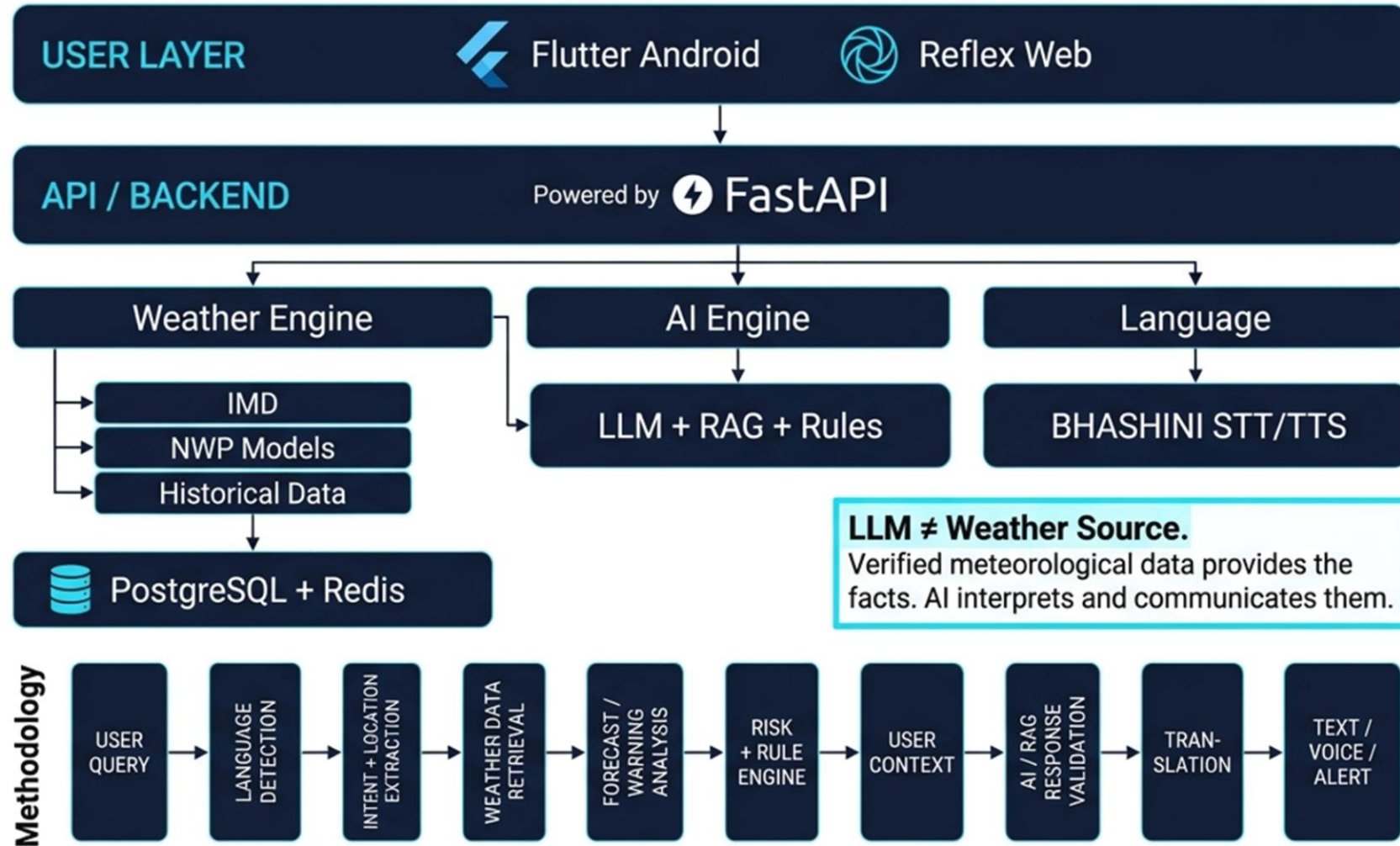
An AI-powered, multilingual weather platform that integrates real-time meteorological data, NWP forecasts, warnings and location intelligence to provide personalized, actionable weather insights through text and voice.

WeatherGPT



What makes WeatherGPT different?





Technology Stack

Frontend

- Flutter
- Reflex
- Leaflet

Backend

- Python, FastAPI
- PostgreSQL
- Redis

AI

- LLM
- RAG
- Intent classification
- Rule-based weather risk engine

Meteorological Data

- IMD APIs
- Open-Meteo/NWP
- GFS/ECMWF model data
- Historical weather datasets

Language

- BHASHINI
- Speech-to-Text
- Translation
- Text-to-Speech

Deployment

- Docker
- GitHub
- Render / Vercel

01 — Data Reliability

Different weather sources may disagree.



Solution

Source prioritization + timestamps + model metadata + confidence indicators.

02 — AI Hallucination

LLM may generate incorrect weather information.



Solution

LLM never generates raw weather facts.
Verified weather data → rule engine
→ structured context → LLM.

03 — Multilingual Accuracy

Weather terminology can be difficult to translate.



Solution

BHASHINI + controlled weather vocabulary + response validation.

04 — Connectivity / API Failure

Weather services may become temporarily unavailable.



Solution

Caching + provider fallback + last-known data indicator + graceful degradation.

Feasibility & Risk Mitigation

[Feasible MVP → Scalable Platform](#)

Technical Feasibility



DATA AVAILABILITY

IMD APIs + NWP forecast models + open gov datasets.



MODULAR AI LAYER

LLM provider swappable without touching core weather engine.



LANGUAGE API

BHASHINI engine handles multilingual speech & translation.



SCALABLE LOCATION

GPS coordinates + PIN code lookup + reverse geocoding.



CLOUD DEPLOYMENT

Containerized FastAPI backend with light frontend clients.

01 Data Reliability

Weather sources may conflict or disagree.

✓ Source prioritization, timestamps & confidence indicators.

03 Multilingual Accuracy

Local meteorological terminology translation loss.

✓ BHASHINI API + controlled weather dictionary & validation.

04 API / Network Failure

External weather services temporarily offline.

✓ Multi-level caching + provider fallback & graceful degradation.

Key Risk Matrix

IDENTIFIED RISK	SEVERITY	MITIGATION STRATEGY
API Outage & Network Loss	HIGH	Multi-layer caching + provider fallback mechanism
AI Hallucination	HIGH	Grounding with rule engine & structured weather context
High User Traffic Spikes	HIGH	Redis caching layer + horizontally scalable FastAPI microservices
Data Disagreement / Translation Error	MEDIUM	Source prioritization + controlled vocabulary dictionaries

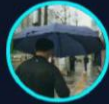
Feasibility Summary: The MVP relies primarily on open/public meteorological datasets, modular APIs, and open-source technologies, keeping initial development cost low while preserving a clear path toward national-scale deployment.

EMPOWERING DIVERSE CITIZEN PERSONAS



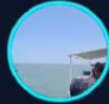
Farmers

Weather-aware agricultural decisions



Students

Travel and daily school planning



Fishermen

Marine & severe weather warnings



Drivers

Rain, wind & visibility awareness



Workers

Heatwave & rain safety alerts



Disaster Mgrs

Location warnings & situational awareness



**BETTER WEATHER
ACCESS**



**BETTER
UNDERSTANDING**



**FASTER DECISION
MAKING**



**REDUCED WEATHER
RISK**



**SOCIAL + ECONOMIC
BENEFIT**

Social Benefits

- Democratizes access to meteorology
- Supports regional-language users (22 languages)
- Improves accessibility through voice input
- Makes technical forecasts clear & simple
- Helps users respond faster to severe weather

Economic Benefits

- Saves agricultural yield & crop investments
- Optimizes urban logistics & delivery routes
- Protects outdoor labor & workforce hours
- Enhances tourism, fisheries & transport planning
- Lowers local disaster response expenditure

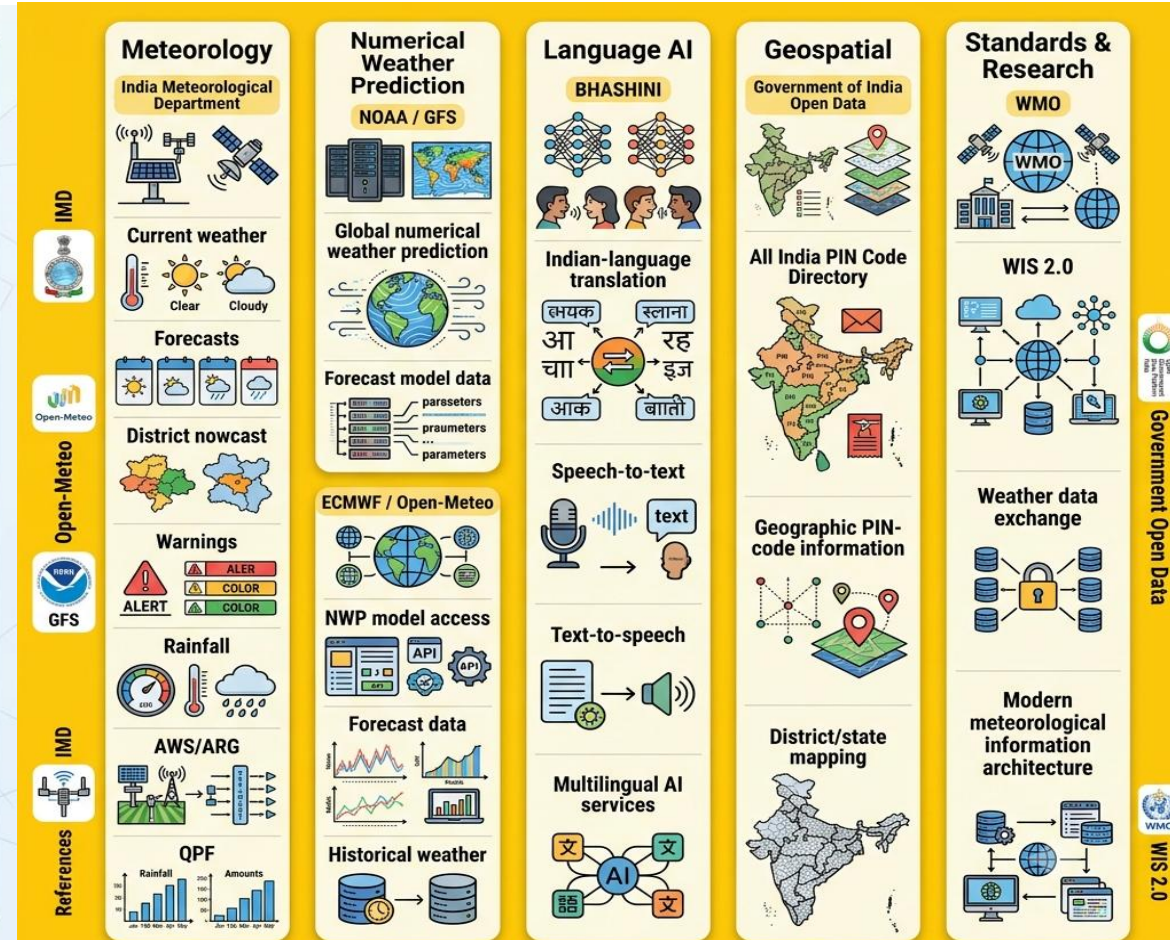
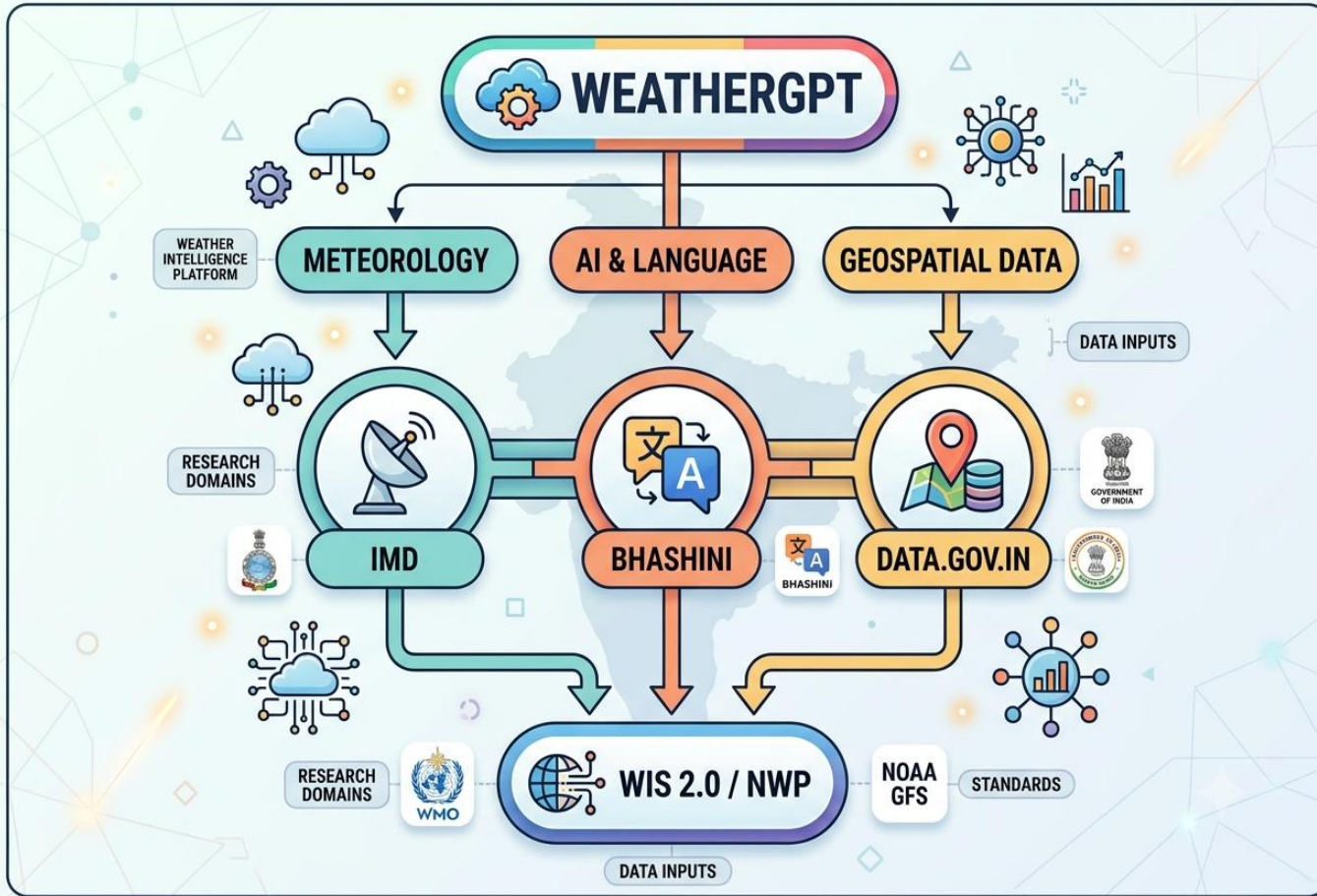
Environmental Benefits

- Weather-informed agricultural practices
- Reduces unnecessary water & chemical usage
- Improves local community climate preparedness
- Builds grassroots climate trend understanding

SCALING HORIZON: ASSISTANT → NATIONAL WEATHER INTELLIGENCE INTERFACE

Citizen → Community → District → State → National

Key Distinction: WeatherGPT does not replace meteorological agencies — it makes their information more **understandable**, **accessible** and **actionable** for citizens.



IMPORTANT INSTRUCTIONS



Please ensure below pointers are met while submitting the Idea PPT:

1. Kindly keep the maximum slides limit up to six **(6)**. (Including the title slide)
2. Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures
3. Keep your explanation precise and easy to understand
4. Idea should be unique and novel.
5. You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).
6. You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.

Note - You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.